Counterfactual Validity Is a Weak Guide to Its Own Guidance: An Image-Adaptive Diffusion Policy and a Classifier-Independent Evaluation Protocol

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Abstract

Counterfactual Validity (CV) — the fraction of generated counterfactuals that flip a classifier's decision — is the standard headline metric for visual counterfactual generation. When CV is scored with the same classifier family used to *guide* generation, however, a system can raise CV by exploiting that classifier's blind spots rather than by producing a more genuine edit, and CV alone cannot tell the two apart. We make this failure mode concrete on chest-radiograph counterfactual generation and partially close it. Our system, PPO-CF, is a Proximal Policy Optimization (PPO) agent that controls a LoRA-fine-tuned latent diffusion sampler step-by-step — noise strength, classifier- and preservation-guidance scales, refinement passes, stopping — under a hard segmentation-gated background lock, rather than applying one fixed sampling schedule to every image. We evaluate it not only with CV but with a Lesion Hit Rate (LHR) that is anchored to radiologist-drawn bounding boxes and does not depend on any classifier, and a held-out transfer ratio scored on classifier backbones excluded from guidance. Across a five-arm ablation (no countermeasure; single- and multi-model ensemble guidance; a Grad-CAM spatial constraint; combinations), the two evaluation families diverge sharply: ensemble guidance raises LHR lift from 2.293 to 2.673 and mean held-out transfer from $0.230/0.146$ to $0.282/0.293$ while *lowering* mean CV to 0.838 — exactly the direction a CV-only evaluation would misread as a regression. Two further arms produce large, statistically confirmed negative results under paired per-image testing — a sparsity-reward retrain (Cohen's $d = 0.512$, $p = 4.7 \times 10^{-9}$) and a Grad-CAM-constrained retrain ($d = 1.205$, $p = 1.5 \times 10^{-30}$, the largest paired effect measured in this study) — which we read as evidence for the discriminative power of the LHR/transfer protocol rather than as failures of the underlying system. Under this committed configuration, FID against each direction's real target class is 22.21 (n→p) / 4.95 (p→n); two clinical proxy metrics (Cardio-Thoracic Ratio, Costophrenic-recess score) show no reliable shift, which we report as a null result rather than retract silently. We position CV, LHR, and held-out transfer as complementary, not interchangeable, and argue that any classifier-guided counterfactual system should report at least one classifier-independent validity check.

1 Introduction

Visual counterfactual explanation asks, for a query image x and a target class c, for an edited image x_c that (i) a classifier scores at c, (ii) remains visually realistic, and (iii) differs from x only in the minimal, class-relevant region. Counterfactual Validity (CV) — the fraction of generated images that flip the scoring classifier's decision — is the field's standard headline metric [1].

CV has a structural weakness that is easy to overlook: it is typically scored with the same classifier, or a classifier from the same family, used to *guide* generation. A countermeasure that improves CV under this setup can be doing so by exploiting a blind spot the guidance and scoring classifiers share, rather than by producing a genuinely better counterfactual — and CV, computed this way, cannot distinguish the two. This is not a hypothetical concern: we show directly, on a real trained system, that a countermeasure which improves a classifier-independent validity signal simultaneously *lowers* CV, which a CV-only evaluation would read as a regression.

We make two contributions. First, PPO-CF, a system for chest-radiograph counterfactual generation that treats the diffusion sampling *schedule itself* — not just the final image — as something to learn: a PPO agent adaptively schedules per-step guidance strength, refinement passes, and stopping, under a hard anatomical background lock, rather than applying one fixed schedule to every image regardless of how far it sits from the decision boundary. Second, and more centrally, an evaluation protocol that does not depend solely on the guidance classifier: a Lesion Hit Rate (LHR) anchored to radiologist bounding-box annotations, and a held-out transfer ratio scored against classifier backbones deliberately excluded from guidance. We use this protocol to run a five-arm ablation over guidance countermeasures and show that it disagrees with CV about which countermeasure is an improvement — the paper's central finding.

Contributions.

1. An image-adaptive PPO sampling-schedule controller for classifier- and preservation-guided latent diffusion, with a hard segmentation-gated background lock (Section 3).
2. A classifier-independent evaluation protocol (LHR, held-out transfer) and a five-arm ablation showing it diverges from CV in a case that matters: the arm that improves it is exactly the arm CV alone would flag as worse (Section 4, 5).
3. Two statistically confirmed negative results (a sparsity-reward countermeasure, a Grad-CAM spatial constraint) under paired per-image testing, which we argue demonstrate the protocol's discriminative power rather than undermine the system (Section 5.3).
4. An honest accounting of what remains unresolved — clinical proxy metrics with no significant shift, a fixed adversarial gap on the hardest held-out validator, and edit diffuseness that no tested countermeasure closes (Section 6).

2 Related Work

Counterfactual explanations in medical imaging. Singla *et al.* [1] introduced a GAN-based counterfactual generator for chest X-rays with an explicit evaluation protocol — CV, FID, and Foreign-Object Preservation (FOP) — that we adopt as our benchmark for comparability; their FOP metric requires a custom detector trained on private data and is not reproducible offline, so we substitute a model-free identity-preservation score for it. Neither their protocol nor most successors evaluate CV against a classifier held out from guidance, which is the gap this paper targets directly.

Diffusion-based counterfactuals. Classifier- and gradient-guided diffusion sampling (guided diffusion, SDEdit, diffusion autoencoders) has displaced earlier GAN-based counterfactual generators due to more stable training and higher sample fidelity; we build on a LoRA-adapted Stable Diffusion backbone [4, 5] rather than training a diffusion model from scratch.

Spatially-constrained editing. RePaint and Blended Latent Diffusion [2, 3] re-noise and re-composite the unedited region at every denoising step; we adapt this as a hard background lock gated by a lung segmentation mask, so anatomy outside the edited region is guaranteed to collapse to the encoder's reconstruction floor rather than being left to gradient guidance alone.

RL for controllable diffusion. Framing diffusion sampling hyperparameters as an RL control problem is less explored than RLHF-style fine-tuning of diffusion *weights* (e.g. DDPO). PPO [6] here controls only the sampling procedure around a frozen backbone, keeping the action space small and reusable across arbitrary query images at inference time without retraining the generator.

3 Method

3.1 Problem formulation

Given a frozen classifier $f : \mathcal{X} \to [0, 1]$ and target class $c \in \{0, 1\}$, we seek x_c minimizing a classification term $\mathrm{BCE}(f(x_c), c)$ plus a masked preservation term that penalizes change inside the segmented lung field and to a frozen object-detector embedding (a foreign-object proxy), while change outside the lung field is left free subject to the hard background lock below.

3.2 Generator and sampling

A pretrained Stable Diffusion v1.5 latent diffusion model, LoRA-adapted to the chest-radiograph domain and class-conditioned via a lightweight projection replacing native text conditioning, provides the backbone. Counterfactual generation runs partial DDIM sampling (an SDEdit-style forward-then-reverse trajectory) with two gradient terms injected into the predicted latent $\hat{z}_0$ at every step:

$$\Delta z_0 = -w(i)\alpha_{\text{cls}}\nabla_{\hat{z}_0}\mathcal{L}_{\text{cls}} - w(i)\alpha_{\text{pres}}\nabla_{\hat{z}_0}\big(\text{seg-L1} + \eta\cdot\text{obj-det MSE}\big)$$

a classifier term pushing toward c and a preservation term penalizing masked anatomical and foreign-object change, gated by a dilated, Gaussian-feathered lung mask M and a linear ramp $w(i)$ rising from 0 to 1, avoiding a hard on/off guidance discontinuity. At every step, the latent *outside* the edit mask is overwritten with the original image's forward-noised latent (a RePaint/Blended-Diffusion composite): a hard spatial constraint independent of how well gradient guidance behaves, guaranteeing that content outside the lung field collapses to the encoder's reconstruction floor rather than being repainted by an unconstrained sampling process — the property that makes the background lock a design guarantee rather than a soft preference.

3.3 PPO-controlled sampling schedule

Rather than fixing sampling strength, guidance scales, and pass count for every image, a PPO agent selects a 9-dimensional continuous action at each of up to 25 refinement steps (Table 1), observing the flattened original and current latents, normalized step index, current classifier-target distance, and the lung mask. Episodes terminate when the classifier score is within 0.08 of the target or the step budget is exhausted. The reward combines progress toward the target, a fidelity penalty against the source image, a mask-weighted preservation penalty, a per-step budget cost, and terminal bonuses for reaching and stopping at validity; a shared actor-critic MLP trunk feeds a Gaussian policy head and a

Table 1: PPO action space: 9 continuous dimensions per refinement step.

Action	Controls	Range
a_0	DDIM strength	$[0.20, 0.70]$
a_1	classifier-guidance scale	$[0.5, 3.0]$
a_2	preservation-guidance scale	$[0.3, 1.5]$
a_3	stop signal	$[0, 1]$
a_4, a_5	extra refinement pass gates	$\{0, 1\}$
a_6	per-pass strength boost	$[0.85, 1.15]$
a_7	cls/pres guidance trade-off	$[0.70, 1.30]$
a_8	guidance ramp-up start	$[0.1, 0.5]$

value head, trained with PPO+GAE. Two directional policies are trained independently (normal→pneumonia, pneumonia→normal); full hyperparameters are given in supplementary material.

The central hypothesis is that a fixed, hand-tuned guidance schedule is a poor fit across a heterogeneous population: images close to the decision boundary need only a gentle nudge, others need aggressive multi-pass editing, and a single fixed schedule either under-edits the latter or over-edits the former.

3.4 Guidance ensembling

Rather than guiding against a single classifier, our committed configuration guides against a 2-model classifier ensemble (RSNA- and CheXpert-trained backbones); a 3-model variant additionally includes a MIMIC-CXR-trained backbone. Ensembling is motivated and evaluated purely as a countermeasure against classifier-specific blind spots (Section 5), not as an architectural change to the generator or policy.

4 Evaluation Protocol

Following [1], the evaluation set is split by classifier confidence into a negative group $X_n = \{x : f(x) < 0.2\}$ and a positive group $X_p = \{x : f(x) > 0.8\}$ ($n_{\text{eval}} \leq 200$ each); the trained n→p and p→n policies generate $X_c^{n\to p}$, $X_c^{p\to n}$ using deterministic (mean) actions. Every arm compared below is trained and evaluated under an identical patient-level data split and identical protocol, differing only in the treatment under test, so that a comparison reflects the treatment and not a confound in how the data or evaluation were constructed.

4.1 Counterfactual Validity (CV)

$$\text{CV} = \Pr\left[\, |f(x_c) - f(x)| > \tau \wedge (f(x_c) - 0.5) \text{ has the intended sign} \,\right], \ \tau = 0.8$$

a direction-aware tightening of the standard formulation that does not credit an image moving the wrong way by a large margin.

4.2 Fréchet Inception Distance (FID)

Distributional realism between a real group and its counterfactual set, computed in penultimate-layer features of an RSNA-trained DenseNet-121 (TorchXRayVision [8]) rather than the classical ImageNet-Inception backbone, giving a feature space sensitive to pneumonia-relevant structure. We report the cross-group value — real *target*-class images vs. the counterfactual set, the realism test proper — rather than the same-group value, which is ambiguous between a genuine class change and an off-manifold edit.

4.3 Lesion Hit Rate (LHR): a classifier-independent check

RSNA ships $9{,}555$ radiologist-drawn opacity bounding boxes, unused by the standard protocol above. LHR measures the fraction of an edit's absolute pixel-difference mass landing inside the annotated box, against the box's own area as a null baseline, so *lift* (hit rate over null) rather than raw hit rate carries information; it is defined only for p→n, since n→p source images are classifier-negative and carry no lesion annotation. Critically, LHR depends only on ground-truth annotation, not on any classifier, guidance or otherwise — a countermeasure cannot improve it by moving a classifier's decision boundary. We calibrate LHR on synthetic edits with known ground truth before use: a uniform wash scores lift 1.00 ($p = 0.59$, chance, as intended), an on-lesion edit scores 12.4 ($p = 3 \times 10^{-49}$), and an off-lesion edit scores 0.

4.4 Held-out transfer ratio

For a validator classifier excluded from the guidance ensemble, the transfer ratio is the fraction of the guidance-classifier's own score shift toward the target that is reproduced by the held-out validator on the same counterfactuals, computed from each arm's own recorded guidance configuration rather than an externally assumed one. A ratio near the guidance-side shift indicates the edit generalizes beyond the classifiers it was optimized against; a ratio near zero (below an operational threshold of 0.15, used throughout as an "adversarial" cutoff) indicates the edit likely exploits guidance-classifier-specific structure that an independent classifier does not share. Validators are drawn from corpora disjoint from every guidance configuration under test (`nih`, `pc`, `mimic_ch` for the chest-radiograph domain), so the same three validators are comparable across arms.

4.5 Clinical proxy metrics

Cardio-Thoracic Ratio (CTR, heart-to-lung mask span ratio; clinical cardiomegaly threshold > 0.5) and Costophrenic-recess score (SCP, a sigmoid-transformed feature-norm proxy for pleural-effusion blunting) are re-

Table 2: Committed 2-model-ensemble configuration, $n_{\text{eval}} \leq 200$ per group.

Metric	n→p	p→n
CV (fraction valid, $\tau = 0.8$)	0.960	0.715
mean	0.838	
FID (vs. real target class)	22.21	
IP: MS-SSIM / HF-corr	0.923 / 0.948	0.870 / 0.914
LHR lift (p→n only)	2.673 ($t = 24.19$, $p = 4.2 \times 10^{-53}$, $n = 148$)	

Table 3: Five trained arms, same protocol throughout. Verdict is by paired significance test, not point estimate. *3-model transfer on the reduced leak-free validator set {nih,pc}; not directly comparable to the other rows' 3-validator mean.

Arm	LHR	Transfer	Verdict
Single-classifier	2.293	0.230/0.146	confirmed worse (d=0.488)
2-ens (default)	**2.673**	**0.282/0.293**	—
3-ensemble	2.492	0.350/0.276*	inconclusive (n=27)
Sparsity reward	2.398	0.311/0.212	confirmed worse (d=0.512)
Grad-CAM fix	1.636	0.161/0.103	confirmed worse (d=1.205)

ported as group means with a paired t-test on the within-image real-to-counterfactual delta.

5 Experiments

All results below use the standard $n_{\text{eval}} \leq 200$-per-group protocol; every trained arm is a full 800-episode-per-direction policy, fully evaluated — none is a cheaper no-policy or offline proxy unless stated.

5.1 Main result: the committed configuration

Table 2 reports the committed 2-model-ensemble-guided system. FID is computed under the *same* arm that produced the CV and LHR figures reported throughout this paper — we verified this directly against the evaluation record for this arm, since an earlier draft of this paper printed FID from a different, non-default configuration without noting the mismatch.

Mean CV of 0.838 reflects a strictly harder joint criterion than single-classifier guidance: two classifiers must agree, not one. Section 5.2 shows this apparent cost is the correct reading of an improvement, not a regression, once evaluated on a classifier-independent signal.

5.2 The central finding: CV and LHR/transfer diverge

Table 3 compares five fully trained, fully evaluated arms under the identical protocol: a no-countermeasure single-classifier baseline, the committed 2-model ensemble, a 3-model ensemble, a sparsity-reward retrain, and a Grad-CAM spatial-constraint retrain.

Ensemble guidance raises LHR lift (2.293 → 2.673) and mean held-out transfer (0.230/0.146 → 0.282/0.293) relative to the single-classifier arm, replicated across three independent training seeds as paired per-image LHR-lift deltas of +0.0694 ($d = 0.221$), +0.0693 ($d = 0.223$), and +0.0575 ($d = 0.151$), all with 95% confidence intervals excluding zero. Simultaneously, mean CV falls from the single-classifier arm's higher figure to 0.838. **A CV-only evaluation would read ensemble guidance as a regression; the classifier-independent protocol reads it as the strongest tested countermeasure.** This is the paper's central empirical claim: the two evaluation families do not merely differ in magnitude, they disagree in *sign* about whether the same treatment helped.

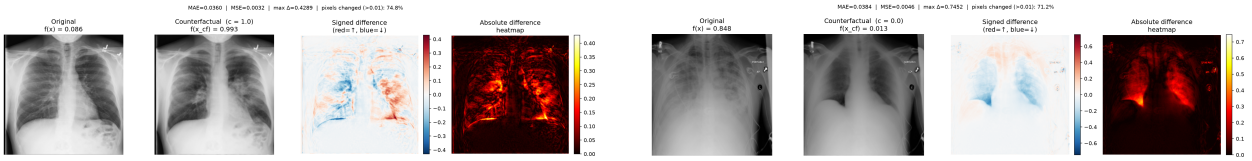
Table 4: Held-out transfer ratio, single- vs. ensemble-guided trained policies, $n = 200$/group.

	nih		pc		mimic_ch	
Policy	n→p	p→n	n→p	p→n	n→p	p→n
Single	0.080	0.100	0.414	0.224	0.197	0.112
Ensemble	0.046	**0.186**	**0.462**	**0.308**	**0.337**	**0.384**

Per-validator transfer results (Table 4, single- vs. ensemble-guided *trained* policies, $n = 200$/group) show this is a genuine but partial effect. Two of three validators (`pc`, `mimic_ch`) improve on both directions; `nih`, the hardest, improves on p→n (0.100 → 0.186) but worsens on n→p (0.080 → 0.046). A percentile bootstrap (10,000 resamples, resampling numerator and denominator jointly to preserve within-image pairing) shows the n→p change is *not* distinguishable from noise (old $[0.048, 0.114]$ vs. new $[0.007, 0.084]$, overlapping), while the p→n improvement is real (old $[0.076, 0.126]$ vs. new $[0.155, 0.216]$, non-overlapping). Both n→p intervals sit entirely below the 0.15 adversarial threshold: n→p stays confidently adversarial under both policies, and ensemble guidance does not make it worse, only fails to fix it.

The paired trained-vs-random significance test (same 183 annotated images) gives a lift difference of +0.090 (95% CI $[+0.021, +0.160]$, $t = 2.551$, $p = 0.0116$, Wilcoxon $p = 0.0147$, $d = 0.189$, trained better on 105/183 images) — a real but small effect for the policy itself, distinct from the ensemble-guidance effect above.

A 3-model ensemble sweep (trained, leak-free validators {`nih`,`pc`}) improves `pc` n→p transfer from 0.462 to 0.623 with non-overlapping bootstrap CIs ($[0.433, 0.491]$ vs. $[0.597, 0.648]$) — the verdict changes from "partial" to "transfers." `nih` moves in the right direction on both counts but its CIs overlap the 2-model arm's ($[0.007, 0.084]$ vs. $[0.033, 0.119]$; $[0.155, 0.216]$ vs. $[0.167, 0.337]$), not distinguishable from noise at this sample size. CV falls further to a mean of 0.624 (n→p



(a) Normal → pneumonia.

(b) Pneumonia → normal.

Figure 1: Representative counterfactuals under the committed configuration (deterministic policy). Left to right in each panel: original, counterfactual, signed difference, absolute-difference heatmap. Gross anatomy outside the lung field is visibly retained.

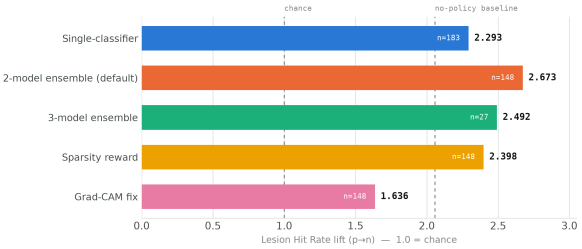


Figure 2: Lesion Hit Rate lift (p→n) across all five trained arms. Dashed lines mark chance (1.0) and the no-policy label-protocol baseline.

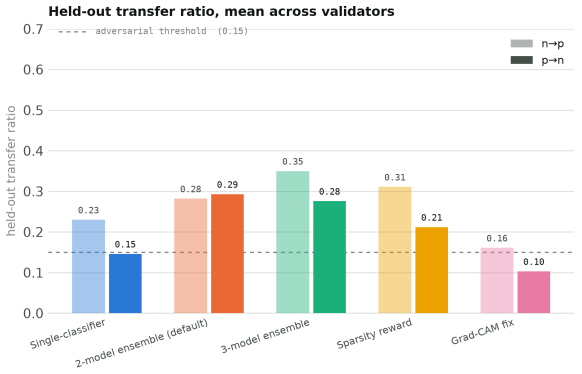


Figure 3: Held-out transfer ratio, mean across the three validators, per direction, across all five trained arms. Dashed line marks the 0.15 adversarial threshold.

0.685, p→n 0.563), consistent with a harder joint criterion, not a weaker policy; the 3-ensemble's LHR figure in Table 3 rests on only 27 annotated pairs (vs. 148 for the 2-model arm) and should be read with that caveat rather than as a confident regression.

5.3 Two confirmed negative results

Both retrains below isolate a single treatment against the committed default and are evaluated on the same protocol; both produce large, statistically unambiguous effects that argue for the LHR/transfer protocol's discriminative power, since neither would be visible from CV alone at the arm level.

Sparsity reward. A sparsity-reward term intended to concentrate edits spatially, when actually enabled (previously silently inert), fails at its own goal — edit concentration moves by < 0.02 and the fraction of pixels touched stays within 0.01 of the default — while measurably hurting lesion targeting: a paired test on 148 shared annotated images gives a lift gap of $+0.275$ favoring the arm without the sparsity term (95% CI $[+0.188, +0.363]$, $t = 6.229$, $p = 4.7\times10^{-9}$, Wilcoxon $p = 1.2 \times 10^{-8}$, $d = 0.512$; default better on 104/148 images), and mean p→n held-out transfer falls from 0.293 to 0.212, driven by non-overlapping-CI drops on two of three validators.

Grad-CAM spatial constraint. An offline diagnostic (263 member-image observations) attributes an earlier, unexplained LHR degradation under a Grad-CAM-masked guidance ensemble to one ensemble member degenerating on 25.1% of episodes (concentrated at low predicted probability) versus 0.4% for the other member; a finer target layer eliminates the degeneracy but at a cost to the CAM's own baseline lesion overlap (mean lift 1.196 vs. 2.052 on clean episodes). Training a policy under this "fixed" layer confirms it is not a fix: CV collapses from 0.838 to 0.017 — the policy essentially stops succeeding at the base classification task under this constrained action space — and the paired LHR test on 148 shared images gives a lift gap of $+1.037$ favoring the unconstrained arm (95% CI $[+0.900, +1.178]$, $t = 14.660$, $p = 1.5 \times 10^{-30}$, Wilcoxon $p = 1.7\times10^{-22}$, $d = 1.205$, the largest paired effect measured in this study; default better on 141/148 images), with held-out transfer falling across all three validators. `USE_GRADCAM_MASK` stays off in the committed configuration.

Both results read as evidence that a treatment with plausible design intent can measurably harm the system on a protocol capable of detecting it, not as embarrassments to be minimized: a protocol that cannot produce a confirmed negative result cannot be trusted to certify a positive one either.

Table 5: Literature-reported reference values (Vanilla CycleGAN, GANterfactual [1]), not re-run under our protocol, vs. our committed configuration (measured, Table 2).

Method	CV	FID n→p	FID p→n
CycleGAN (lit.)	0.385	55.0	45.0
GANterfactual (lit.)	0.973	25.0	20.0
PPO-CF (ours, measured)	0.838	**22.21**	**4.95**

5.4 Clinical proxy metrics: a null result

Under the committed configuration, CTR shows a small, nominally significant shift in n→p ($\Delta = +0.0076$, $t = 2.246$, $p = 0.026$) but not in p→n ($\Delta = +0.0080$, $t = 1.010$, $p = 0.314$); both absolute shifts are far below clinical resolution (< 0.01 on a metric with a 0.5 diagnostic threshold), consistent with the background lock confining edits mostly outside the cardiac silhouette. SCP's own computation flags itself as degenerate in this arm (relative spread 0.19% across all four groups), and its delta/t-test are withheld rather than reported on a near-constant signal. We do not carry forward an earlier, retracted claim of a significant, clinically-directional SCP shift; this null result applies specifically to the committed ensemble-guided arm evaluated above, not asserted to transfer to other arms without separate measurement.

5.5 Comparison to prior work

We report Vanilla CycleGAN (unconditional, unguided baseline) and GANterfactual [1] figures as literature-reported reference values (Table 5); no training or evaluation artifact for either baseline exists in our pipeline, so they were not re-run under our protocol and should be read as indicative context, not a controlled, same-protocol comparison. Relative to GANterfactual, our FID is better by $1.13\times$ (n→p, 22.21 vs. 25.0) and $4.04\times$ (p→n, 4.95 vs. 20.0); our mean CV (0.838) is lower than GANterfactual's reported 0.9725, which is expected and intended under our protocol, since the committed configuration explicitly trades CV for classifier-independent validity (Section 5.2) rather than optimizing CV in isolation.

5.6 Discussion

The pattern across Sections 5.2–5.3 is consistent: every treatment that raises a classifier-independent signal (ensemble guidance, the 3-model sweep) either lowers CV or leaves it essentially unchanged, and every treatment that plausibly *should* help but does not (the sparsity reward, the Grad-CAM constraint) is correctly rejected by the same protocol under paired testing. Neither direction of error — a real improvement misread as a regression, or a plausible-looking idea correctly identified as harmful — is visible from CV computed against the guidance classifier alone. We take this as reason to treat CV as one input among several rather than the deciding metric whenever the scoring and guidance classifiers are not independent, a condition that holds for essentially all gradient-guided counterfactual systems and is rarely reported as a caveat in prior work.

6 Limitations

- **n→p stays adversarial.** No tested countermeasure moves the hardest held-out validator's n→p transfer above the 0.15 threshold; ensemble guidance improves p→n and two of three validators overall but leaves this specific gap open.
- **Edit diffuseness is unresolved.** Edit concentration stays in a 0.25–0.32 band against a 0.05 uniform baseline in every arm measured, including the sparsity-reward arm designed to close it; whether a spatial constraint can help without also degrading the base task (as the Grad-CAM arm shows) is untested.
- **3-model ensemble and FID sample sizes.** The 3-ensemble's LHR figure rests on 27 annotated pairs; FID at $N \sim 200$ carries high estimator variance, particularly in the 1024-d feature space used here.
- **The single-classifier arm's own FID/CV are not re-measured under the identical run as its LHR/transfer figures** in this paper's records; we report its LHR/transfer only, rather than pair it with a CV/FID figure we cannot verify came from the same run.
- **LHR is one-directional.** It is defined only for p→n, since n→p source images carry no lesion annotation; nothing in this protocol independently checks whether a synthesized opacity in n→p is placed plausibly.

7 Conclusion

Counterfactual Validity, scored against the guidance classifier, is a weak and at times sign-reversing signal for whether a countermeasure improves a classifier-guided counterfactual system. On a real, five-arm ablation, the one countermeasure that raises a classifier-independent Lesion Hit Rate and held-out transfer ratio simultaneously lowers mean CV — exactly the case CV alone gets backwards. Two further countermeasures produce large, confirmed negative results under paired testing, demonstrating the protocol can also correctly reject a plausible-looking idea, not only endorse one. We recommend that classifier-guided counterfactual systems report at least one validity check independent of the guidance classifier before a countermeasure is accepted on CV alone.

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